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$$f(x) = e^x \sin x$$

$$f'(x) = e^x \sin x + e^x \cos x = e^x (\sin x + \cos x)$$

$$f'(x) = 0 \Leftrightarrow e^x = 0 \vee \sin x + \cos x = 0$$

e^x :

$e^x = 0$ kan niet, e^x raakt nooit $y = 0$.

$\sin x + \cos x$:

$$\sin x + \cos x = 0$$

$$\Leftrightarrow \sin x = -\cos x$$

$$\Leftrightarrow \frac{\sin x}{\cos x} = -1$$

$$\Leftrightarrow \tan x = -1$$

$$\Leftrightarrow x = -\frac{\pi}{4} + k\pi$$

We checken waar de minima en maxima zitten:

$-\frac{\pi}{4}$:

$$-\frac{\pi}{2} < -\frac{\pi}{4} \Rightarrow f'\left(-\frac{\pi}{2}\right) = -e^{-\frac{\pi}{2}} < 0$$

$$0 > -\frac{\pi}{4} \Rightarrow f'(0) = 1 > 0$$

x	$-\infty$	$-\frac{\pi}{2}$			$-\frac{\pi}{4}$		0	$+\infty$
$f'(x)$		-	-	-	0	+	+	+
$f(x)$		$\searrow$	$\searrow$	$\searrow$	m	$\nearrow$	$\nearrow$	$\nearrow$

$x = -\frac{\pi}{4} + k2\pi$ zijn lokale minima

$\frac{3\pi}{4}$:

$$0 < \frac{3\pi}{4} \Rightarrow f'(0) = 1 > 0$$

$$\pi > \frac{3\pi}{4} \Rightarrow f'(\pi) = -e^\pi < 0$$

x	$-\infty$	0			$\frac{3\pi}{4}$	π			$+\infty$
$f'(x)$		+	+	+	0	-	-	-	
$f(x)$		$\nearrow$	$\nearrow$	$\nearrow$	M	$\searrow$	$\searrow$	$\searrow$	

$x = \frac{3\pi}{4} + k2\pi$ zijn lokale maxima

References